MATH3881/5881: Statistical Machine Learning Theory

Week 2: Tutorial Problems — Theory of Feedforward Neural Networks

Sarat Moka *UNSW, Sydney*

These two problems accompany the first half of the Week 2 lecture notes (*Theory of Feedforward Neural Networks*). They cover the role of *hidden layers* and the *Universal Approximation Theorem* for ReLU networks. Attempt them with pen and paper before the tutorial; a separate solutions document is provided a week after the tutorial session.

Problem 1: Why we need hidden layers, XOR

The **XOR** function maps the four binary inputs to the labels

$$(0,0) \mapsto 0, \qquad (0,1) \mapsto 1, \qquad (1,0) \mapsto 1, \qquad (1,1) \mapsto 0.$$

(a) Consider a *linear* classifier $\hat{c}(x) = \mathbf{1}\{w_1 x_1 + w_2 x_2 + b > 0\}$ for $w_1, w_2, b \in \mathbb{R}$. Show that *no* choice of (w_1, w_2, b) can correctly label all four XOR points. (*Hint:* plot the four points in the plane and observe that the two "1" points cannot be separated from the two "0" points by any single straight line.)

(b) Now consider a one-hidden-layer ReLU network with $N_1 = 2$ hidden units and identity output activation, with parameters

$$W^{[1]} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad b^{[1]} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad W^{[2]} = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad b^{[2]} = 0.$$

For each of the four binary inputs $x \in \{0,1\}^2$, compute the hidden activations $a^{[1]} = \sigma_{\text{ReLU}}(W^{[1]}x + b^{[1]})$ and the output $f_\theta(x) = W^{[2]}a^{[1]} + b^{[2]}$. Verify that the network reproduces XOR exactly.

Problem 2: Why the UAT works for ReLU networks

Consider a one-hidden-layer ReLU network with 1-D input and 1-D output:

$$f_\theta(x) \;=\; \sum_{i=1}^{N} \alpha_i \, \sigma_{\text{ReLU}}(w_i x + b_i) \;+\; c,$$

where $w_i, b_i, \alpha_i, c \in \mathbb{R}$ and $w_i \neq 0$. The architecture is shown in Figure 2.1.

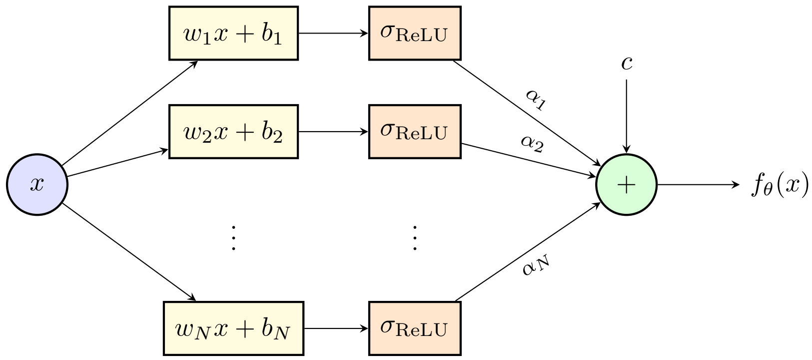

Figure 2.1: Architecture of the one-hidden-layer ReLU network in Problem 2: the input x feeds N parallel affine transformations $w_i x + b_i$, each followed by a ReLU activation; the resulting activations are then combined by a weighted sum (weights α_i) and an output bias c to give $f_\theta(x) = \sum_{i=1}^{N} \alpha_i \, \sigma_{\mathrm{ReLU}}(w_i x + b_i) + c$.

(a) Show that, for any choice of parameters, f_θ is a *piecewise-linear* function of x with at most $N + 1$ linear pieces on $\mathbb{R}$.

(*Hint:* start by sketching what a single shifted ReLU term $\alpha_i \, \sigma_{\mathrm{ReLU}}(w_i x + b_i)$ looks like as a function of x, and identify where it changes slope.)

(b) A standard result from analysis says that any continuous function $f^\star$ on a compact interval $[a, b]$ can be approximated uniformly to arbitrary accuracy by a piecewise-linear function with sufficiently many pieces (just interpolate $f^\star$ at a fine enough grid and connect the points by straight lines). Combine this fact with (a) to write a short, informal argument for the following statement: *as $N \to \infty$, one-hidden-layer ReLU networks can approximate any continuous function $f^\star : [a, b] \to \mathbb{R}$ uniformly to arbitrary accuracy.*

Note. The statement in part (b) is the univariate Universal Approximation Theorem for ReLU networks. The structural fact in (a), *a ReLU network is just a piecewise-linear function*, makes the theorem feel almost obvious once you see it.